

W-09 Using Non-Traditional Data to Nowcast Pacific Island Trade

A Bridge Equation Approach with AIS and Alternative Data

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The Case for Nowcasting Pacific Islands Trade

Why trade nowcasting matters globally and why the Pacific Islands are a particularly compelling case.

QUESTION 1

Why Nowcast Trade?

Early warning signals

Trade identifies turning points in global economic activity, demonstrated clearly during COVID-19

Ripple effects

Disruptions anywhere create large supply chain cascades (e.g., Red Sea, Panama Canal disruptions)

Publication lag

Official trade data lag 2–9 months; AIS-based nowcasts available within just 7 working days

QUESTION 2

Why Pacific Islands Specifically?

Import dependence

Imports average ~60% of GDP, making trade economically critical.

Data scarcity

Over half of Pacific island countries publish trade data with a lag of 12+ months.

Fiji as regional hub

Fiji is the South Pacific transshipment hub; declines in port traffic signal broader regional slowdowns early

Example of the Impact of Iran-US war

How a Gulf shipping shock would transmit into Fiji's import bill and the policy decisions it would inform.

Illustrative transmission channel:

1 Gulf conflict & shipping risk

Escalation around the Strait of Hormuz lifts war-risk insurance and re-routes tankers.

2 Oil & freight repricing

Brent and marine bunker fuel reprice within days. Charter rates and insurance premia follow.

3 Fiji's import bill

Fiji imports nearly all its fuel, so the bill rises on unit values first. Higher freight also lifts the landed cost of every other import.

4 Domestic pass-through

Pump prices, bus fares, power tariffs and inter-island freight rise; food and building materials follow.

Policy questions this raises

Monetary policy

The RBF must read core against headline before moving the policy rate and a nowcast of the import shock is an input to that judgement.

Fiscal policy

Excise relief, transport subsidies and targeted transfers each carry a budget cost that has to be sized against how long the shock is expected to last.

The common constraint: every one of these decisions needs the size of the import shock now but Fiji's official merchandise trade statistics arrive two to three months later.

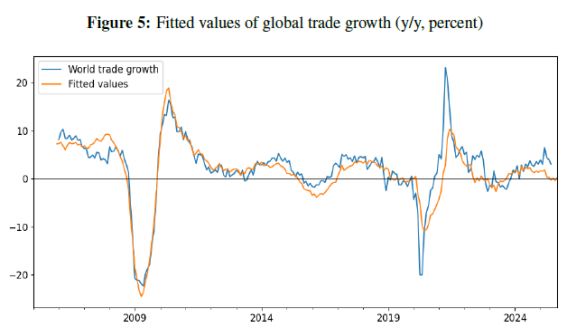
Key prior works

Two key prior studies on trade nowcasting and AIS data, and the gap this project addresses.

Nowcasting World Trade with a Multi-Region Factor Model

Jackson & Rivera Greenwood (2026) · IMF Working Paper WP/26/048

- **Main contribution:** A dynamic factor model with a block structure: one global factor plus regional factors and cross-region spillovers.
- **Literature Gap**: The analysis is limited to global trade aggregates and Pacific Island economies fall outside its scope.

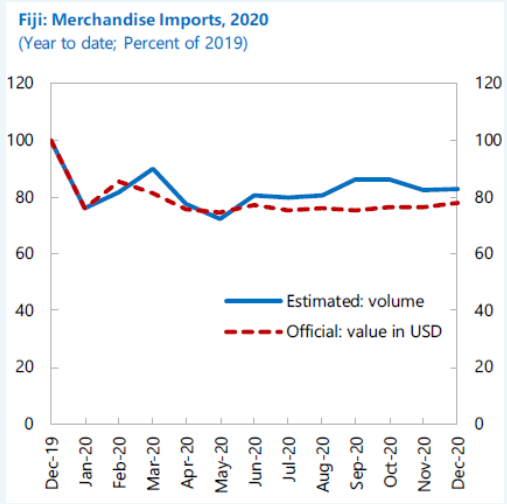


Sources: Author's estimates.

Tracking Trade from Space: An Application to Pacific Island Countries

Arslanalp, Koepke & Verschuur (2021) · IMF Working Paper WP/21/225

- **Main contribution**: estimate cargo by watching how much a ship's draft (how deep it sits in the water) changes between arrival and departures.
- **Literature Gap:** Instead of Nowcasting pacific trade, it constructs a proxy measurement of trade activities using AIS signals.



Sources: National authorities and authors' estimates.

What is it and why it matters

What AIS data captures and the three core shipping variables built from raw AIS data.

What is AIS Data: Automatic Identification System (AIS) is fundamentally a safety system, not a statistics system. Vessels self-report their identity and its mandatory for all commercial ships over 300 gross tons and any passenger ships.

1

Voyage-related information

Ship location, speed, course, heading, rate of turn, destination, draft, and estimated arrival time.

2

Static Information

Ship ID, ship type, ship size and dimensions

3

Dynamic Information

Positional aspects (latitude and longitude)

Three core variables

Port calls

Count of vessel arrivals at a port on a given day, also split by vessel type: container, dry bulk, tanker, general cargo and RoRo.

Incoming shipments

Estimated cargo discharged, in metric tons. Inferred from the change in a vessel's reported draft between arrival and departure.

Outgoing shipments

Estimated cargo loaded, in metric tons, the same draft-difference logic in reverse. Published as export volume.

Download AIS Data from IMF Port Watch here

Building the candidate indicator set

The candidate indicators considered, organized by data frequency and by five thematic blocks.

Candidate Data Frequency:

Daily
AIS port calls & shipments · oil price · exchange rate

→

Weekly
Chokepoint transits · commodity and freight trackers

→

Monthly
Visitor arrivals · CPI · private credit · reserves

→

Monthly — the target
Official merchandise imports

Five candidate blocks:

1

Trade & Maritime Activity

PortWatch port calls and incoming / outgoing shipments by vessel type; container throughput; partner-country mirror trade statistics.

2

External Sector & Real Economy

Visitor arrivals and tourism earnings; remittances; electricity generation; VAT turnover; cement sales; GDP components.

3

Financial Sector & Prices

Private sector credit; broad money; lending rates; CPI and its fuel and transport components; FJD exchange rate; reserves and import cover.

4

Regional, Global & Trading Partner

Australia, New Zealand and China industrial production and retail sales; world trade volume; global oil, food and freight price indices.

5

Alternatives: Google Trends, AIS & Dummies

Search intensity for travel and shipping terms; AIS-derived vessel indicators; dated dummies for border closures, cyclones and policy changes.

Choosing between competing specifications

How AIC, adjusted R^2, and out-of-sample RMSE guided the choice between competing specifications.

Golden rules

1. AIC
2. Adjusted R^2
3. Out-of-sample RMSE

Candidate specifications

✅ **Model (1)**

Dependent Variable: IM_SA
Method: Least Squares
Date: 07/16/26 Time: 14:00
Sample (adjusted): 2020M03 2026M03
Included observations: 73 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.023765	0.022581	1.052399	0.2966
IM_SA(-12)	-0.302758	0.052997	-5.712758	0.0000
RES_SDR_USD_SA(-2)	0.059144	0.025244	2.342898	0.0223
AIC_INC_CONTAINER_SA_I(-2)	0.014004	0.006440	2.174498	0.0334
CRISIS_DUMMY	-0.169447	0.076049	-2.228132	0.0294
LOAN_TRANSPORT_SA	0.907935	0.110894	8.187400	0.0000
REM_MNO_SA	-0.075929	0.010514	-7.221820	0.0000
RECOVERY_DUMMY	0.207641	0.040773	5.092612	0.0000
D(REER_IMF(-1))	0.023216	0.012939	1.794270	0.0775

R-squared	0.885728	Mean dependent var	0.097623
Adjusted R-squared	0.871444	S.D. dependent var	0.317096
S.E. of regression	0.113694	Akaike info criterion	-1.395615
Sum squared resid	0.827283	Schwarz criterion	-1.113230
Log likelihood	59.93996	Hannan-Quinn criter.	-1.283080
F-statistic	62.00836	Durbin-Watson stat	1.880099
Prob(F-statistic)	0.000000		

v.s.

❌ **Model (2)**

Dependent Variable: IM_SA
Method: Least Squares
Date: 07/16/26 Time: 13:56
Sample (adjusted): 2020M03 2026M03
Included observations: 73 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.669497	0.467619	-1.431715	0.1571
IM_SA(-12)	-0.274150	0.054590	-5.022007	0.0000
RES_SDR_USD_SA(-2)	0.061518	0.025335	2.428118	0.0180
AIC_INC_CONTAINER_SA_I(-2)	0.014312	0.006491	2.204788	0.0311
CRISIS_DUMMY	-0.156189	0.076514	-2.041320	0.0453
LOAN_TRANSPORT_SA	0.826630	0.119544	6.914860	0.0000
REM_MNO_SA	-0.071807	0.010841	-6.623351	0.0000
RECOVERY_DUMMY	0.211521	0.041133	5.142303	0.0000
REER_RBF(-1)	0.006576	0.004455	1.475976	0.1449

R-squared	0.883931	Mean dependent var	0.097623
Adjusted R-squared	0.869422	S.D. dependent var	0.317096
S.E. of regression	0.114585	Akaike info criterion	-1.380009
Sum squared resid	0.840295	Schwarz criterion	-1.097624
Log likelihood	59.37033	Hannan-Quinn criter.	-1.267474
F-statistic	60.92427	Durbin-Watson stat	1.905118
Prob(F-statistic)	0.000000		

Transformations, timing and shocks

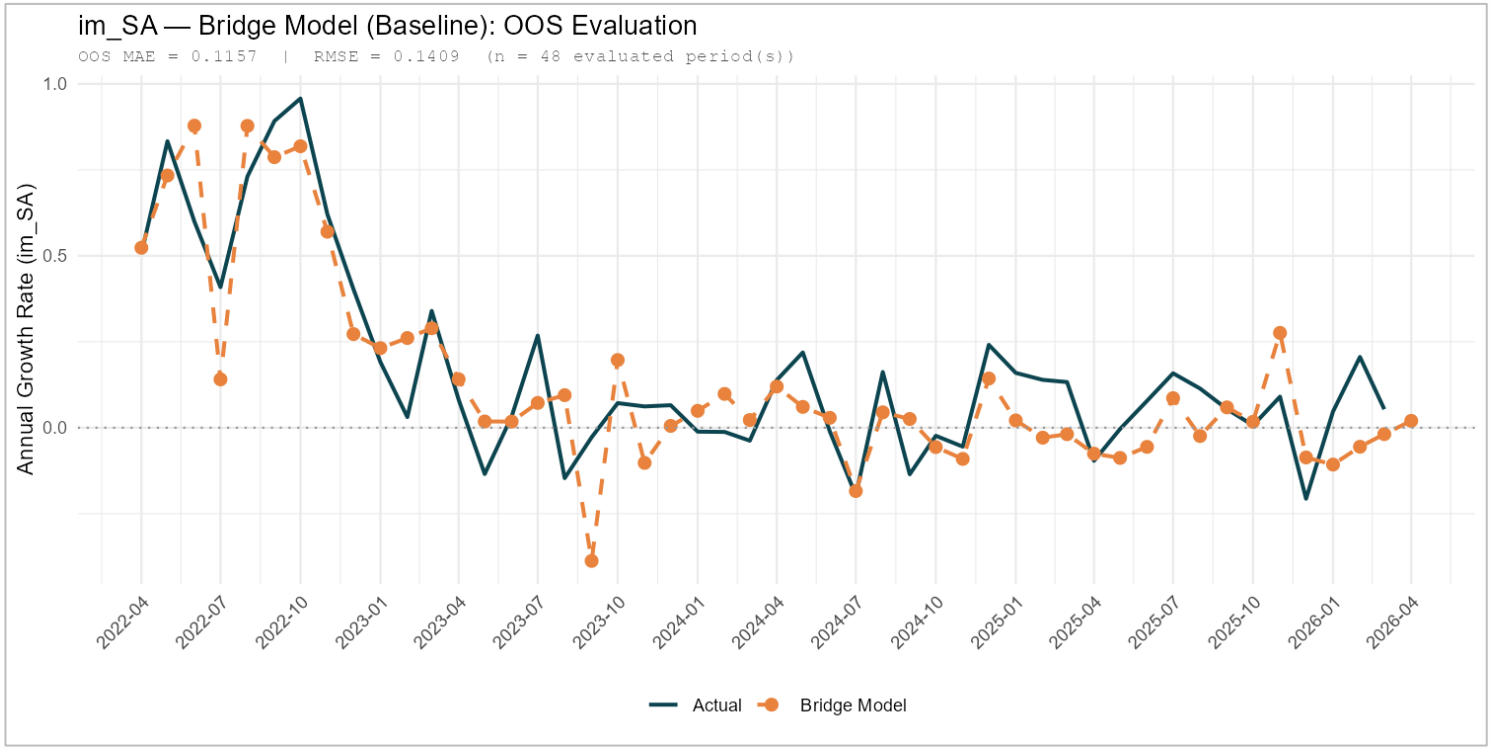
The key modeling choices on data transformation, seasonal adjustment, frequency conversion, and event dummies.

1	**Data transformation**	Work in logs to stabilize variance and test each series for a unit root (ADF, KPSS) before deciding between levels and differences.
2	**Seasonal adjustment**	Fiji trade and tourism are strongly seasonal. Adjust with X-13ARIMA-SEATS and apply it consistently.
3	**Frequency conversion**	High-frequency indicators are aggregated to the target frequency by using sum, average, or last conversion methods.
4	**Dummy variables**	Reserve dummies for identifiable, dated events: a border closure, a natural disasters (cyclones). One dummy per unusual observation fits history.

Out-of-sample evaluation (2022M4~2026M3)

Out-of-sample forecast accuracy of the nowcasting model, measured by RMSE, MAE, and Theil's U2.

Out-of-sample Evaluation Chart



Out-of-sample Evaluation Metrics

Root Mean Square Error (RMSE)
0.12

Mean Absolute Error (MAE)
0.14

Theil U2
0.73

CONCLUSION

A weekly, AIS-based signal for Fiji's import activity

The nowcasting story in four steps, from the data-lag problem to the results and next steps.

The nowcasting story, in four steps:

1 The problem

Official merchandise trade statistics arrive two to three months after the reference period, too slow for real-time policy calls.

2 The data

IMF PortWatch AIS feeds track port calls and estimated cargo tonnage at Suva, Lautoka and four smaller ports, refreshed weekly.

3 The model

A parsimonious bridge equation maps five screened, mixed-frequency indicator blocks onto monthly merchandise imports.

4 The result

Out-of-sample (2022M4–2026M3): RMSE 14.1%, MAE 11.6%, Theil's U2 0.73, beating a naive random-walk benchmark.

What this delivers, and where it goes next

For policymakers

A weekly, AIS-based read on Fiji's import bill that RBF and the Ministry of Finance can use between official trade releases.

Next steps

Extend this nowcasting exercises with more nowcasting models, including MIDAS, PCA, 3PRF, Dynamic Factor Model, and Machine Learning.

Bottom line: Space-based AIS data close the Pacific Islands' trade data gap without waiting on customs processing.

Thank You!